Income Distribution - Problem Set 2

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Question 1 - Variables

You will need to use or construct a variable for hourly wages, and use variables for demographics. In particular, you will need to use information on Total Income, After-tax income, Wages and Salaries, Hours worked, Education, Age, and Gender, as well as the survey weights. List the codes of these variables for your two data sets. Use variables for individuals, not families. Note: Since we will be focussing on wages in this assignment, you do not need to consider business income (differently from the first assignment).

Answer:

Table 1: Variable Name Mapping for SCF and CIS Datasets

Variable Description	**SCF Label (1997)**	**CIS Label**
Total Income	TOTINC	TTINC
After-tax Income	INCAFTTX(INCFTX)	ATINC
Wages and Salaries	WAGSAL	WGSAL
Hours Worked	HRSWRK(USHOURS)	ALHRWK
Education	EDUC(EDUCA)	HLEV2G
Age	AGE	AGEGP
Gender	SEX	SEX
Survey Weights	WEIGHT	FWEIGHT

Question 2 - Building hourly wages

(a) Download the consumer price index for the relevant years from Statistics Canada. Plot its evolution. (b) Compute hourly wages for each individual. Deflate the hourly

wage using the CPI (use Statscan's annual CPI). To make wage statistics more informative for us, use 2020 as the base year. Hint: maybe the most straightforward way of doing this is to do it year by year in your do file, with one line of code per year.

Answer:

(a)

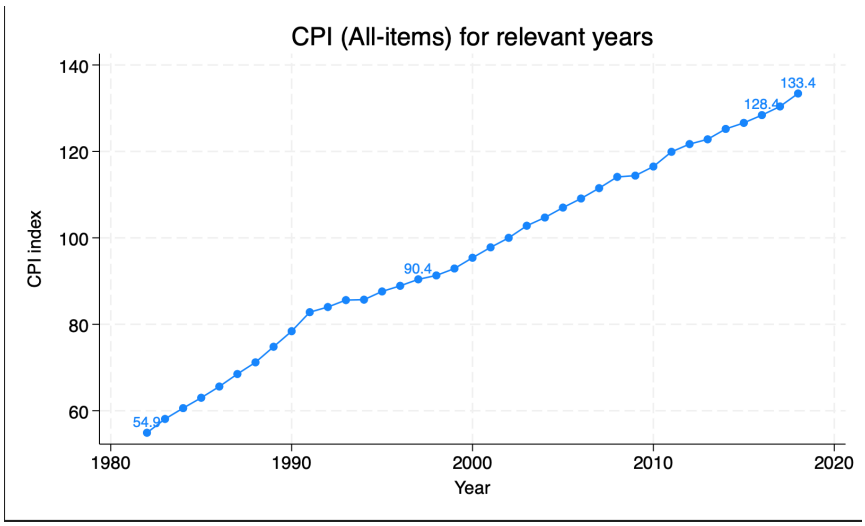

Figure 1: CPI Evolution

Question 3 - Sample selection

Drop from your sample individuals with missing information, zero hours, or a wage below 50% of the minimum wage. Next, keep only individuals between the ages of 25 and 60 (inclusive), who work more than 260 hours. Report in a table like Table 1 of HPV how many observations you have (not using weights) at each stage. Use this selected sample for the remainder of the assignment. Hint: use the yearly minimum wages I provide on mycourses as an Excel file. In principle, you would need to be more precise here, since minimum wages are set at the provincial level in Canada. But to keep things simple, you can use the provided numbers for all individuals. (The file contains the unweighted average of the prevailing minimum wage in Alberta, BC, Ontario and Québec, in current dollars.) Hint 2: In some surveys, age is not

available, only a variable for year of birth, in 5-year intervals. I ask you to use a sample of individuals aged 25 to 60. I suggest that if most individuals in an age group fall in the sample, you keep that age group. For example, those born in 1968 are 25 years old in 1993. You could then include the group of all of those born in 1965-1969.

Answer:

Table 2: Sample selection in the SFC and CIS

	SFC 1982		SFC 1997		CIS 2016		CIS 2018	
	Dropped	Remaining	Dropped	Remaining	Dropped	Remaining	Dropped	Remaining
Initial Sample		79,517		69,461		62,414		94,336
Missing info	41,688	37,829	36,048	33,413	31,287	31,127	47,674	46,662
Wage < 0.5 × min. wage	3,598	34,231	2,642	30,771	1,387	29,740	1,785	44,877
Sample A		**34,231**		**30,771**		**29,740**		**44,877**
Age 25–60	8,285	25,946	4,365	26,406	7,725	22,015	12,562	32,315
Sample B		**25,946**		**26,406**		**22,015**		**32,315**
Annual hours > 260	169	25,777	117	26,289	487	21,528	750	31,565
Sample C		**25,777**		**26,289**		**21,528**		**31,565**

Question 4 - The evolution of inequality

(a) Plot the time series of the log variance of total income, after-tax income, and wages and salaries (similar to HPV, F11). Plot, in a different graph, the time series of the Gini coefficient of these variables. (b) Plot, in a single graph, the time series of the log variances of annual wages, hourly wages, and annual hours, as well as the correlation of log hours and log wages. Repeat this for men, and for women. (c) Plot the time series of percentiles 5, 50 and 95 of the distribution of hourly wages. A "time series" here consists of values for the four years you are analyzing. +4

Answer:

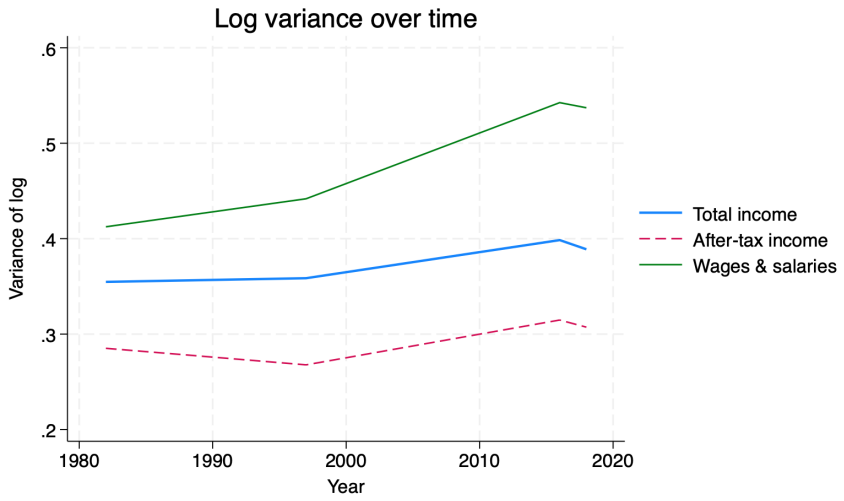

Figure 2: Graph 1 - (a)

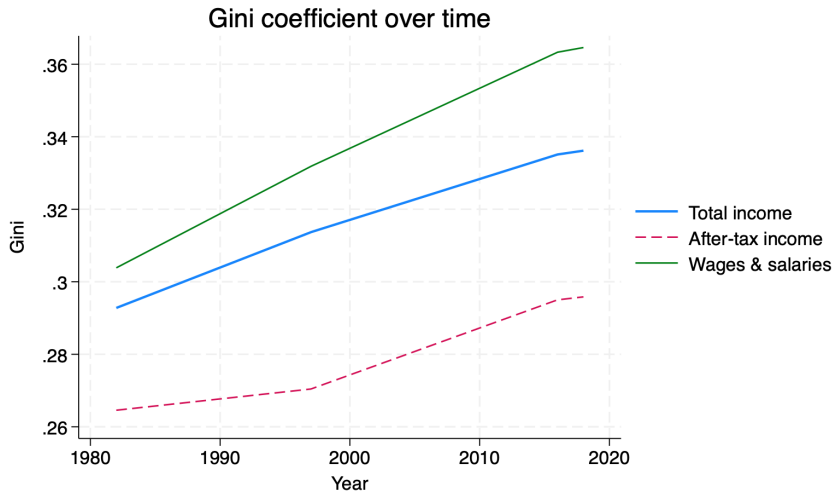

Figure 3: Graph 2 - (a)

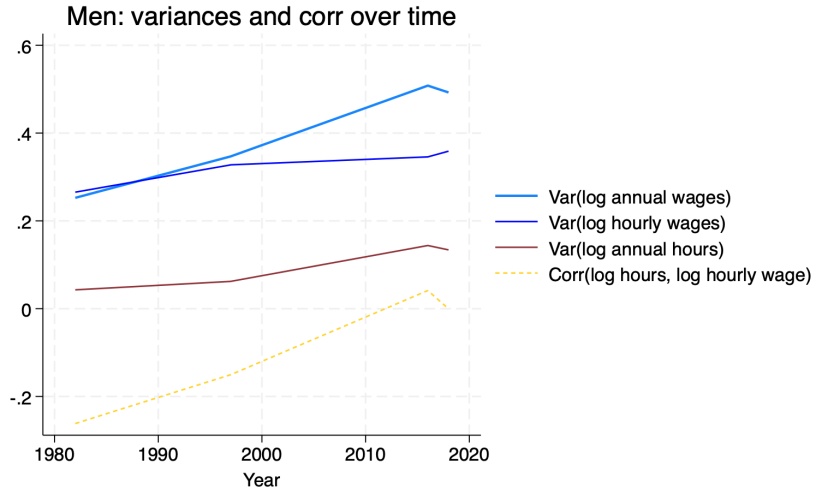

Figure 4: Graph 3 - (b)

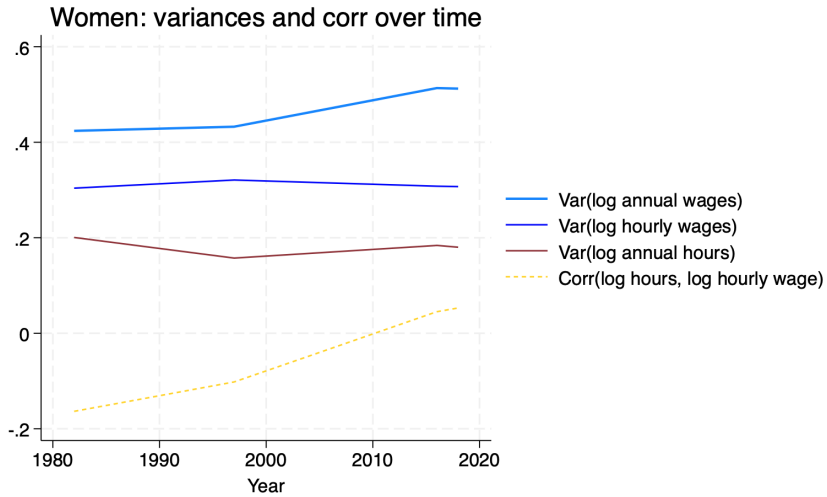

Figure 5: Graph 4 - (b)

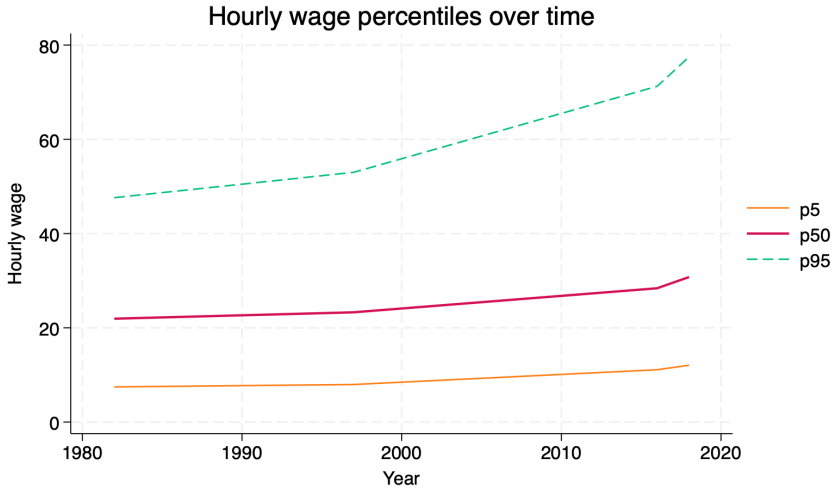

Figure 6: Graph 5 - (c)

Question 5 - Composition-adjusted wages

Here, you will compute composition-adjusted wages, as in AKK. (a) Generate a variable potential experience as age minus years of schooling minus 6. You may need to create a new variable years of schooling based on a categorical schooling variable for this. (b) Create a variable group distinguishing gender education potential experience groups. How many groups do you have? Hint: use the egen... group function. Hint 2: You need to make sure that your groups are identical across your different data sources. (c) Compute the share of the sample (using weights!) in each group in each year. (Show only the code for this.) (d) Compute the mean wage in each group and year. (Show only the code for this.) (e) For each year, compute the composition-corrected mean wage for (a) the entire population, (b) women, (c) men, (d) university graduates, and (e) high-school graduates, as discussed in the lecture on AKK. Report naive and composition-corrected wages for each of these five broad groups, for each of your years. (f) Discuss how composition-adjusted wage changes over the sample period compare to naive ones for the five broad groups. Hint: There are several ways of addressing this question. Probably the most elegant option is to use the collapse command in Stata, at the group level. This was not covered in the Stata intro. If you try this, be aware that collapse overwrites your data set in memory, so you need to make sure that you save it first. Another option is to report the group-level information using tab and tab ..., sum, copy it to Excel (note the Copy table function in Stata), and process it there.

Answer:

For (a) I had to reduce to 4 educational groups for 1982 data due to the fact Post-grad is not defined.

As such there are 32 groups, $2 \times 4 \times 4 = 32$.

For (c), 1982 and 1997 have no weights available.

(f)

Over the full sample period, the relationship between naive and composition-adjusted wages changes in a clear way. In the first year (1982), naive wage measures largely reflected price effects, with relatively smaller composition distortions. However, as the workforce evolved—particularly through increased educational levels and changing participation patterns—the gap between naive and composition-adjusted wages widened. By the later years, composition-adjusted wages were consistently above naive wages across most groups, suggesting that broader changes increasingly influenced observed wage averages.

For the overall population, naive wage growth increasingly understates underlying wage levels once composition is held fixed. Educational upgrading expands the share of higher-educated workers. Over time, these compositional forces exert downward pressure on raw averages relative to fixed-weight (AKK) measures. The divergence becomes more pronounced in the later years.

Among women, compositional effects are particularly strong. The rise in educational attainment and labor force participation significantly alter the structure of the female workforce over time. As a result, naive wage measures incorporate both return changes and large compositional shifts. In later years, the adjusted–naive gap is clearly larger than in earlier years, indicating that a significant portion of observed female wage dynamics reflects structural workforce change rather than purely changes in returns.

For men, compositional effects are present but more moderate. Male workforce composition evolves more gradually, so the gap between naive and adjusted wages is smaller throughout the period. Wage movements for men are therefore driven more by price changes than by shifts in observable characteristics. Nonetheless, even for men, compositional forces become somewhat more visible in the later years.

Across education groups, the pattern is consistent with long-run educational expansion. For university graduates, expansion of higher education introduces marginal, relatively lower-paid entrants, causing naive averages to understate underlying wage levels. For high-school graduates, increasing negative selection into the group depresses naive means relative to fixed-composition benchmarks. Overall, by 2016 and

2018, composition effects are clearly shown in observed wage dynamics, and naive measures increasingly distort workforce change with true wage movements.

Note:

***I beg you to make the last assignment shorter, it took maybe 20-25 hours to complete this, during exam week, purely due to the complications that arose at each step.